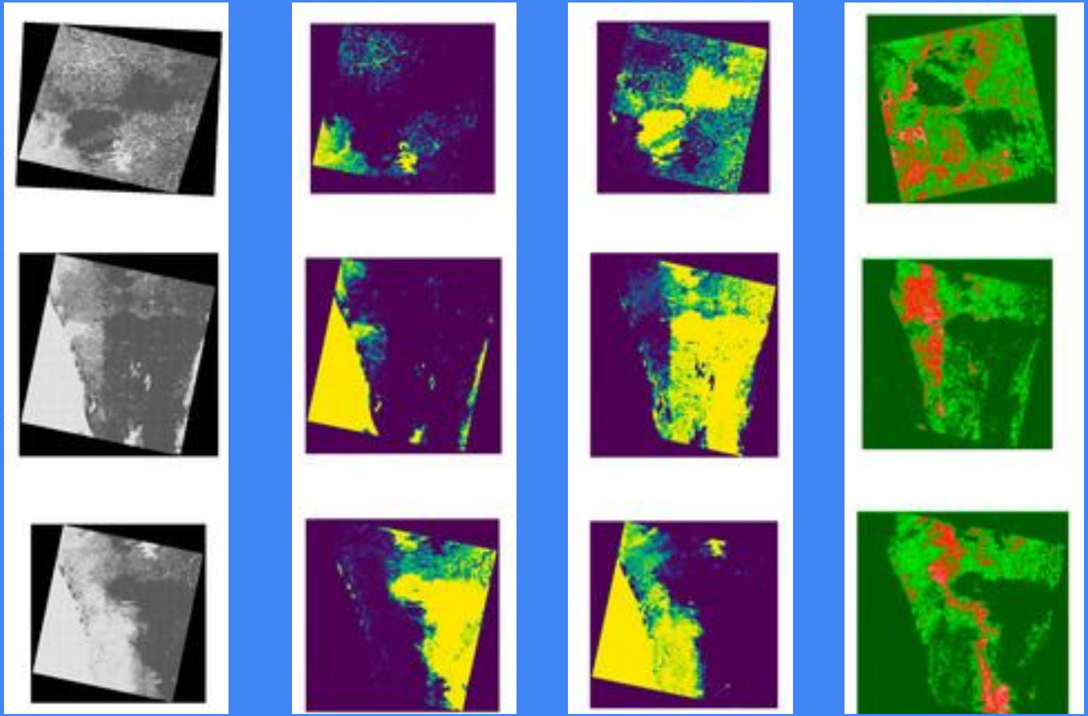


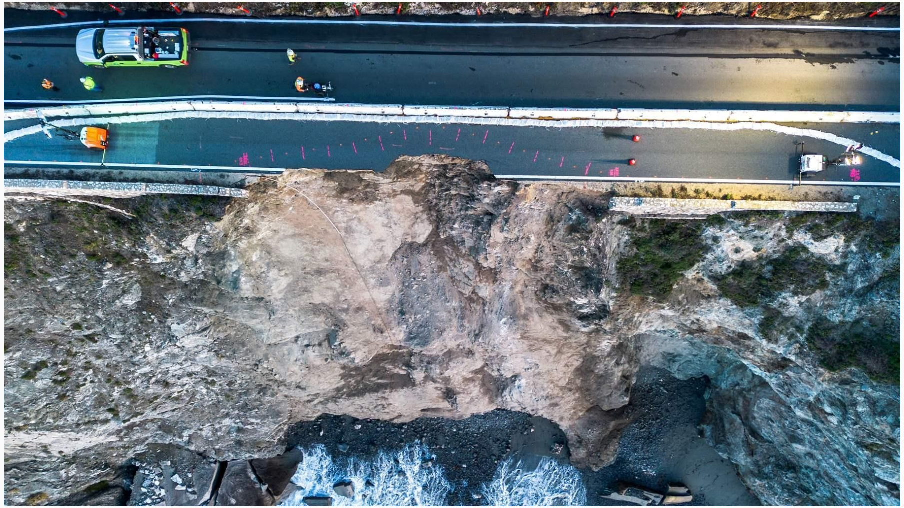
MUSA 650 Final: Remote Sensing Solution for Urban Planning
Flood Mapping for Urban Disruptions and Emergency Response

Shuya Guan and Riya Saini

Objectives

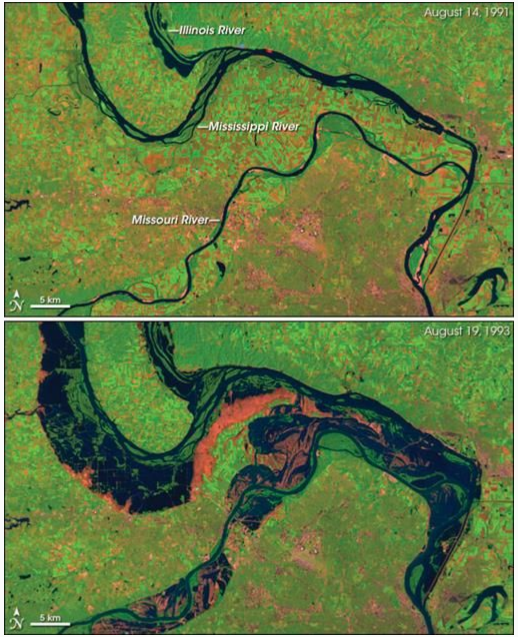
Climate change is leading to increased vulnerability to flooding for urban transportation infrastructure—highways, rail lines, and bridges

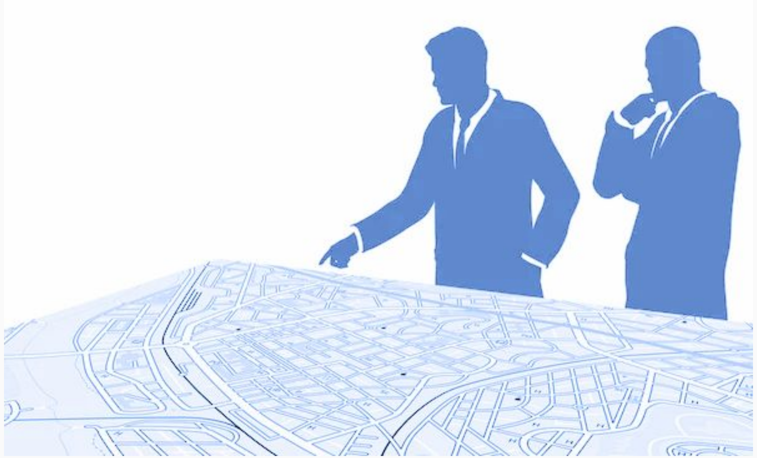
Current flood risk assessments lack the spatial granularity and predictive insight needed to prioritize interventions and plan resilient routes.



Problem Scope

By detecting flood-prone areas, our analysis can be overlaid with critical transportation corridors, to highlight infrastructure segments that are most at risk. These vulnerability maps will empower urban planners and transportation agencies to strategically prioritize climate-resilient investments, plan effective detours, and design adaptive infrastructure solutions.





Target Users

Data Source

The **SEN12-FLOOD Dataset** is built from 336 time series containing **Sentinel 1 and Sentinel 2 images** of areas that suffered a major flooding event during the 2019 winter. The acquisition period goes from December 2018 to May 2019. The observed areas are clustered in East Africa, South West Africa, Middle-East and Australia.

Sentinel 1

Sentinel 2

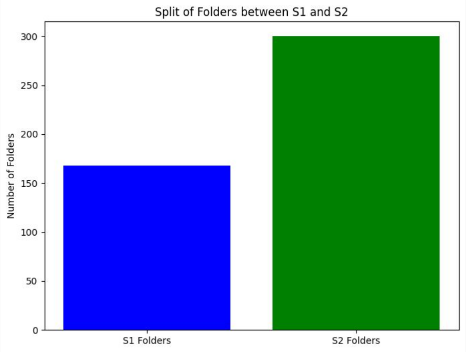
Sentinel 1: grayscale (single-band)

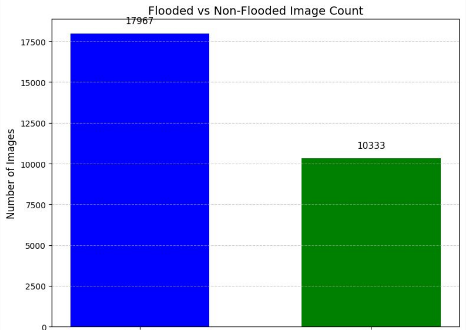
Band name / Polarization	Pixel Size	Wavelength
VV	10 m	5.405GHz
VH	10 m	5.405GHz

Sentinel 2: RGB (three-band color)

Band name	Pixel Size	Wavelength
B1	60 m	443.9nm (S2A) / 442.3nm (S2B)
B2	10 m	496.6nm (S2A) / 492.1nm (S2B)
B3	10 m	560nm (S2A) / 559nm (S2B)
B4	10 m	664.5nm (S2A) / 665nm (S2B)
B5	20 m	703.9nm (S2A) / 703.8nm (S2B)
B6	20 m	740.2nm (S2A) / 739.1nm (S2B)
B7	20 m	782.5nm (S2A) / 779.7nm (S2B)
B8	10 m	835.1nm (S2A) / 833nm (S2B)
B8A	20 m	864.8nm (S2A) / 864nm (S2B)
B9	60 m	945nm (S2A) / 943.2nm (S2B)
B11	20 m	1613.7nm (S2A) / 1610.4nm (S2B)
B12	20 m	2202.4nm (S2A) / 2185.7nm (S2B)

Data Exploration





Why We Use Sentinel-2 RGB Images？

Compatible with deep learning models

Sentinel-2 images can be reconstructed into standard RGB format using bands B04, B03, and B02. This makes them directly suitable for models like CNN and U-Net that expect 3-channel input.

Provides higher spatial resolution and urban detail

Compared to grayscale Sentinel-1 SAR images, Sentinel-2 data offers clearer visual features of flooded areas including roads, textures, and buildings.

Offers a larger and more scalable dataset

We identified approximately 23,975 Sentinel-2 image files. This volume exceeds the available Sentinel-1 images, providing a more robust base for training and evaluation.

Pre-Processing

Steps

- Band Selection (B02, B03, B04)

  B02.TIFF
  B03.TIFF
  B04.TIFF
  → Folder(RGB)

- Reflectance Scaling
- Percentile Normalization (2–98%)
- Cloudy Scene Filtering
- RGB Image Composition (same date)
- Data Augmentation (flip, rotate)
- Folder-level Splitting (train/val/test)

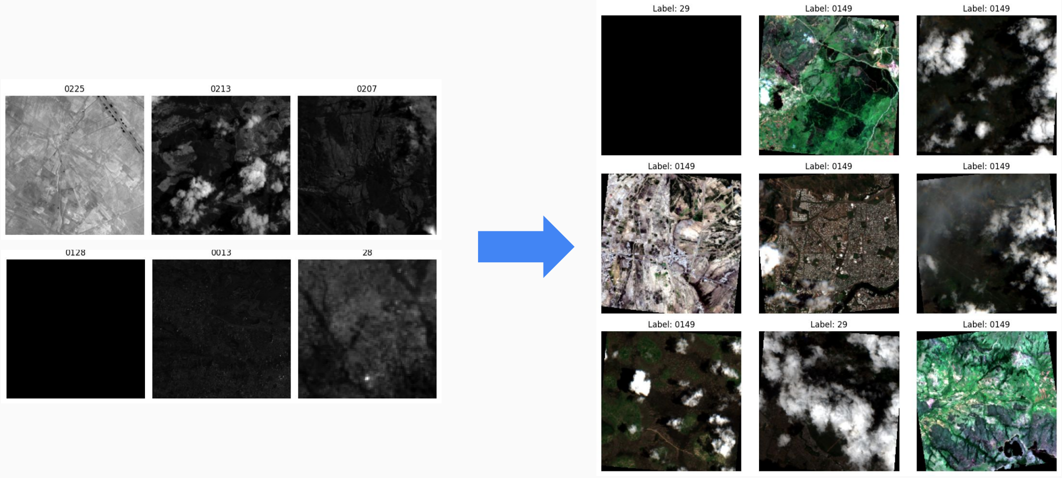
Valid Folders Result (300 / 335)

300 folders confirmed with valid RGB bands captured on the same date.

Train / Val / Test Split

Dataset split by folders:

70% train (210), 15% val (45), 15% test (45) to avoid data leakage.



Model Development

UN SPIDER

UN-SPIDER Workflow United Nations Platform for Space-based Information for Disaster Management and Emergency Response

Model Type: Threshold-based / rule-based with optional ML integration, Uses Sentinel-1 SAR (VV/VH) backscatter before and after the event.

Methodology:

- Pre-event and post-event image comparison
- Apply thresholds (e.g., 2–3 dB difference) to detect permanent water vs flooded areas
- Uses Normalized Difference Flood Index (NDFI) or ratio/mean change detection
- Integrates: Decision trees

Simple classifiers -Random Forests, SVM
Often paired with GIS layers (elevation, land use)

Goal: Simple, operational, fast – especially when computational power is limited in disaster zonesing and Evaluation
Training and Evaluation

WORLDFLOODS

Developed by European Space Agency (ESA), Oxford University, and partners

Model Type: Deep learning-based using CNNs- Focus on multi-class classification e.g., flooded vs. non-flooded vs. permanent water

Methodology:
- Trained on 13 flood events across different countries
- Uses Sentinel-2 multispectral imagery fused with Sentinel-1 SAR for robustness under cloud cover
- U-Net convolutional neural network
- Outputs a pixel-wise binary flood mask
- Can incorporate elevation and slope from DEM data

Supervised learning approach with flood extent as ground truth
Compatible with mobile platforms and satellites with onboard AI,
Designed for real-time or near-real-time flood detection and Works in tandem with GIS tools for post-disaster response mapping

Goal: Provide near-real-time flood maps using open-access satellite data, Prioritize global applicability, even in data-poor or disaster-prone regions, Support humanitarian efforts via integration with UN, NGOs, and local agencies

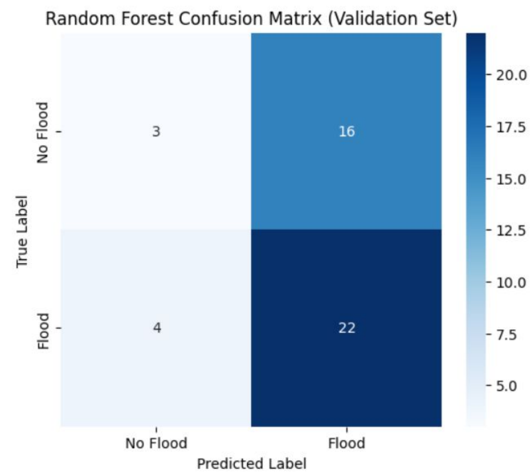
Random Forest

Source: A globally applicable and transferable machine learning approach to flood mapping using Sentinel-1 and Sentinel-2

- Handle high-dimensional, non-linear data well
- Robust to noise and overfitting
- Generalizable results across diverse geographic and hydrological conditions, even with limited labeled data.

```
Cross-Validation Accuracy Scores: [0.57142857 0.52380952 0.61904762 0.71428571 0.69047619]
Mean CV Accuracy: 0.6238 ± 0.0713
```

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', max_depth=15,
                       min_samples_leaf=2, n_estimators=200, n_jobs=-1,
                       random_state=42)
```



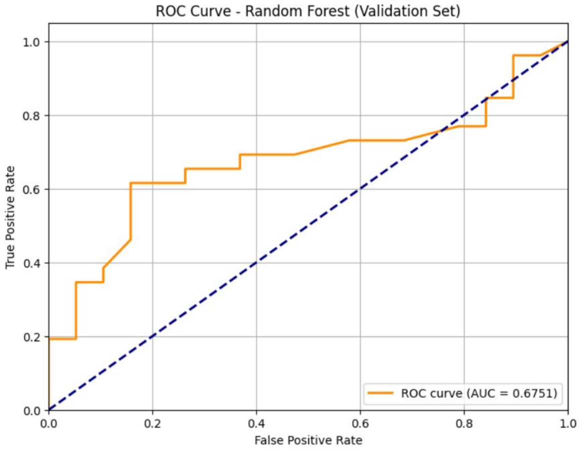


Image-Based Feature Engineering

NDVI (vegetation)

MNDWI (water index)

ResNet CNN

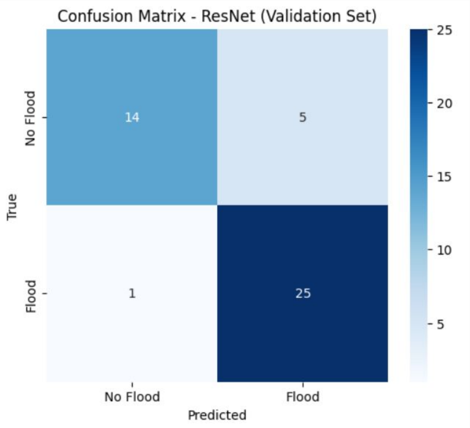
Our model is a transfer learning approach that uses a pretrained ResNet-18 architecture on the ImageNet dataset, modified for binary classification (flooded vs. not flooded)

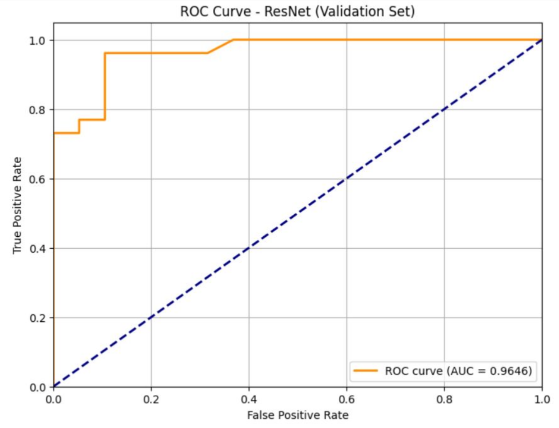
- Uses skip connections to mitigate vanishing gradients and enables deeper learning- helps extract high-level spatial features
- Early layers of ResNet learn generic image features such as edges, textures, and color patterns.
- Improve accuracy on small datasets, Reduce training time

```
Input: RGB Image (3 × 224 × 224)

┌──────────────────────────────────────┐
│           ResNet-18 Backbone         │ (pretrained on ImageNet)
│--------------------------------------│
│ Conv2d(3, 64, kernel_size=7, stride=2, padding=3)  → (64 × 112 × 112)
│ BatchNorm2d(64)                    → (64 × 112 × 112)
│ ReLU                               → (64 × 112 × 112)
│ MaxPool2d(kernel_size=3, stride=2) → (64 × 56 × 56)
│
│ ┌────── BasicBlock × 2 ──────┐      → (64 × 56 × 56)
│ └────── BasicBlock × 2 ──────┘      → (128 × 28 × 28)
│ ┌────── BasicBlock × 2 ──────┐      → (256 × 14 × 14)
│ └────── BasicBlock × 2 ──────┘      → (512 × 7 × 7)
│
│ AdaptiveAvgPool2d((1, 1))           → (512 × 1 × 1)
│ Flatten                             → (512,)
└──────────────────────────────────────┘

→ **Fully Connected Layer (Modified)**
  Linear(512, 2)                      → Output: (batch_size × 2)
```





EfficientNet CNN

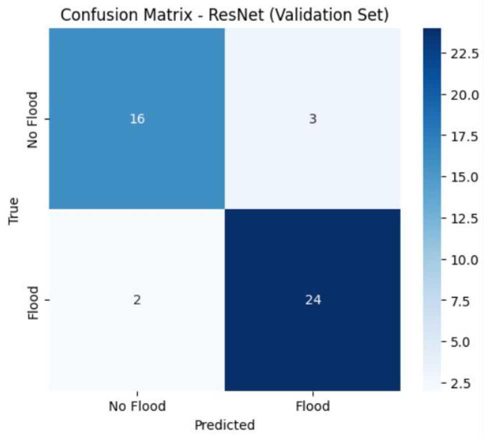
The base EfficientNet-B0 is built upon MobileNetV2's inverted bottleneck residual blocks

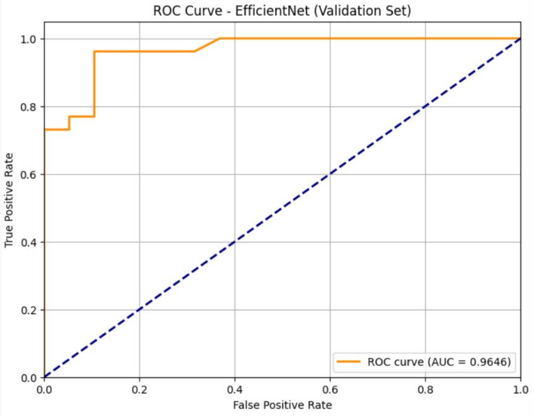
- Employs a compound scaling CNN method that uniformly scales network depth, width, and resolution
- Pretrained on ImageNet
- Final fully connected layer is replaced with a new layer having two output neurons to classify images as 'Flood' or 'No Flood'
- High accuracy with efficient computation

```
Input: RGB Image (3 × 224 × 224)

┌─────────────────────────────────────────────────────────────┐
│                  EfficientNet-B0 (pretrained)                │
│─────────────────────────────────────────────────────────────│
│ Conv2d(3, 32, kernel_size=3, stride=2, padding=1)            │
│ BatchNorm2d(32), Swish, Depthwise Separable Convs            │
│ → MBConv blocks with expansion, squeeze-and-excitation (SE) │
│ → progressive channel increase: 16 → 24 → 40 → 80 → 112 → 192 → 320 │
│ AdaptiveAvgPool2d((1,1))                                     │
│ Flatten                                                     │
└─────────────────────────────────────────────────────────────┘

→ **Modified Classifier:**
  Linear(1280, 2)  # 1280 input features → 2 output classes
```





Next Steps

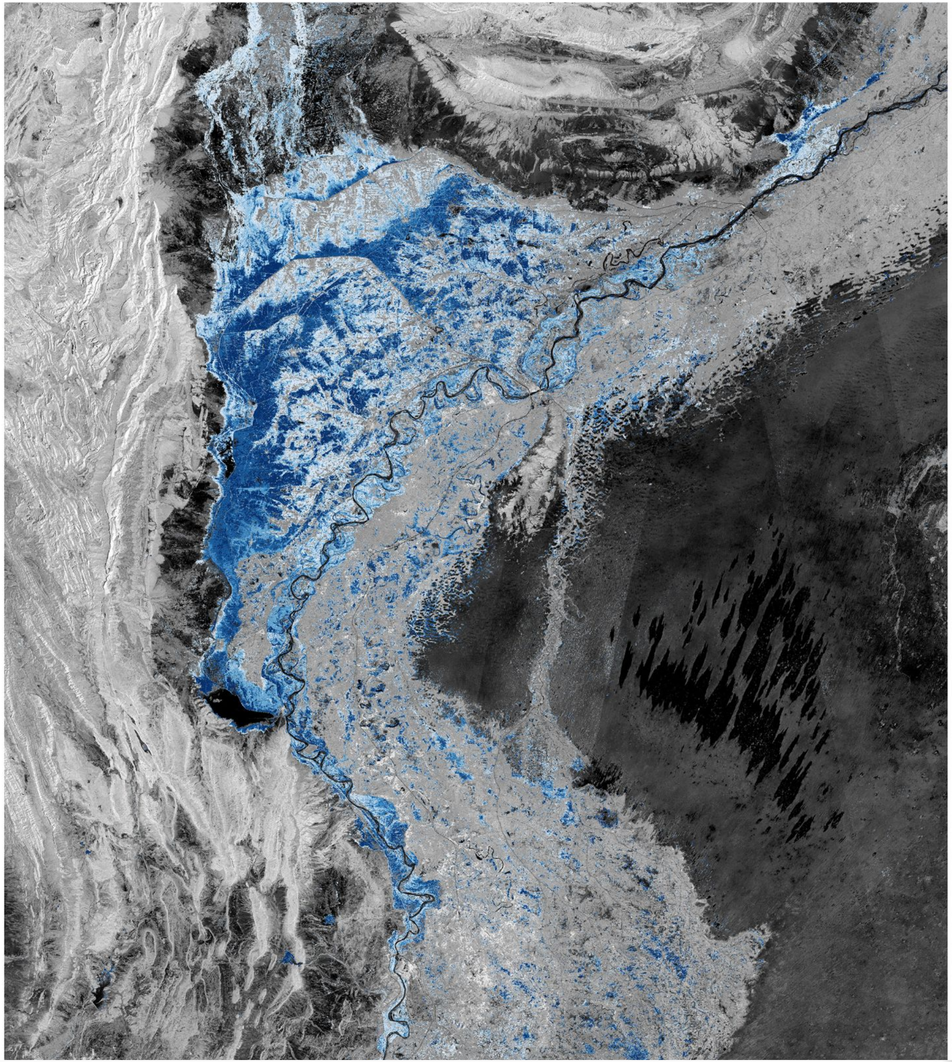
Evaluation

on validation set

- Confusion Matrix
- Accuracy, Precision, Recall, F1 Score, Specificity
- ROC Curve and AUC
- K-Fold cross-validation

Predictions

- Predict labels for test set
- Compute accuracy, precision, etc.
- Plot ROC curve, calculate AUC
- Review misclassifications



Feedback